Scheme Collapse Model: Formula Specification

Core formula, variables, defaults and data sources for the Urdu scheme calculator · Version 0.1 (demo) · 15 October 2026 · Engine file: `formula.js`

Status. This document specifies the model the calculator uses. The recruitment parameters (g, M, k, q) are **placeholders that have not been calibrated** against real collapsed schemes. Results below show how the model behaves. They are not findings. Until a methods note with calibration exists, do not describe the tool as "based on a paper".

1. What the model answers

The user gives three numbers: **the amount** they would put in, **the promised return**, and **how often it pays**. The model answers in three lines:

1. **When** the scheme's money runs out (a calendar month, shown as "at the latest").
2. **How much** a family joining now would lose.
3. **What the same money would earn** in a bank or National Savings product. The calculator asks which bank only after lines 1 and 2 are shown, using a tile for each bank.

2. Core idea

- A scheme promising r per month owes r times the money it holds, every month.
- It has no real income. Payouts can only come from new members' deposits.
- Any city or network has a limited number of people with money, so recruitment slows as the market fills.
- The first month the cash on hand cannot cover what is owed is the **collapse month**.
- The family's own amount does not change the collapse date. It only scales their loss and the bank comparison.

This is a simplified, step-by-step version of the Ponzi dynamics model in Artzrouni (2009). The differences are that this one runs in discrete monthly steps, recruitment is logistic, and payouts are cohort-based.

3. Variables and parameters

3.1 Inputs

Symbol	Name	Unit	Meaning	Source
A	Amount	Rs	Money the family would put in	User
r	Promised return	fraction per period	e.g. 0.05 = 5% per period. For a lump sum, the total gain (1.0 = money doubles)	User
f	Payout frequency	category	daily, weekly, monthly, quarterly, yearly, or lump sum	User
n	Lump-sum period	months	Time until the single payout (lump sum only). Days are converted at 365/12 days per month	User

a	Scheme age	months	How long the scheme has been running (optional; replaces the typical-joiner default)	User, optional
P_{Rs}	Promise in rupees	Rs	Alternative to r: rupees paid per period, or for a lump sum the total paid back. Converted with §4a	User, optional
m_{a}, R_{u}	Already invested	months, Rs	How long ago the family put money in, and (optionally) how much they have received so far (§8b)	User, "already invested" mode
τ	Tax on profit	fraction	Withholding tax for the bank comparison: filer, non-filer, or none (§13)	User
b	Bank rate	fraction per year	Profit rate of the bank product the user picks after seeing lines 1 and 2, before tax	Bank table (§13)

3.2 Model parameters

Symbol	Name	Unit	Meaning	Default
g	Recruitment growth	per month	Growth rate of new money while the market is still empty	0.3
M	Market size	seed units	The most outside money the scheme can ever raise, as a multiple of its starting money	1,000
k	Operators' cut	fraction	Share of each new deposit the operators keep and never pay out	0.1
q	Reinvestment	fraction	Share of each payout members roll back in instead of taking as cash	0
σ	SECP threshold	multiple	If the promised monthly return is at or below $\sigma\times$ the bank's monthly rate, no date is shown	1.5
b_{ref}	Reference savings rate	fraction per year	Savings rate banks show small savers (10%). Used only for the SECP test, so the verdict does not change with the bank picked	10%
H	Horizon	months	Longest simulation before the model gives up	600

3.3 Derived quantities

Symbol	Meaning
e, p	Payout rule: pay $p\times$ balance every e months (§4)
t	Month number since the scheme started ($t=0$ is the seed month)
D_t	New outside money deposited in month t
E_t	Cumulative outside money raised up to and including month t
P_t	Cash pool at the end of month t
B_i, s_i	Balance and start month of cohort i (everyone who joined in the same month)
O_t	Cash owed to all cohorts in month t
C	Collapse month
J	Month the family joins
R, L, Π	Family's cash received, loss, and bank profit over the same months

E^*	Tipping point: money raised beyond which new cash no longer covers payouts

All money inside the simulation is measured in **seed units**: the scheme starts with 1. This makes the collapse month independent of the scheme's real size. Only A is in rupees.

4. Payout schedule conversion

The ledger moves one month at a time, so every promise is converted into "pay p × balance every e months":

User says	e (months)	p	Note
r per day	1	$r \times 365/12$	About 30.4 daily payouts summed into one month
r per week	1	$r \times 365/7/12$	About 4.35 weekly payouts per month
r per month	1	r	
r per quarter	3	r	Paid every third month
r per year	12	r	
gain r after n months	$\lceil n \rceil$	$1 + r$	Paid once, principal included; then the deposit ends. 70 days → month 3

$$\text{owed}_i(t) = p \cdot B_i \quad \text{if } (t - s_i) \text{ is a positive multiple of } e \quad [\text{lump sum: only if } t - s_i = e] \tag{1}$$
$$\text{owed}_i(t) = 0 \quad \text{otherwise}$$

Monthly-equivalent promised rate, used only for the SECP test: $r_m = p/e$ for periodic schedules, and $(1 + r)^{1/e} - 1$ for lump sums.

5. The monthly ledger

Seed (t = 0): $E_0 = 1$, $P_0 = 1 - k$, one cohort with $B = 1$, $s = 0$.

Then for each month t = 1, 2, 3, …

New money (logistic recruitment): $D_t = g \cdot E_{t-1} \cdot (1 - E_{t-1} / M)$ (2)

Money raised: $E_t = E_{t-1} + D_t$ (3)

Pool after deposits: $P'_t = P_{t-1} + (1 - k) \cdot D_t$ and a new cohort opens with $B = D_t$, $s = t$ (4)

Cash due: $O_t = (1 - q) \cdot \Sigma_i \, \text{owed}_i(t)$ (5)

Collapse test: if $P'_t < O_t$ then $C = t$ and stop (6)

Otherwise pay out: $P_t = P'_t - O_t$; rolled-back money $q \cdot \text{owed}_i$ is added to B_i (a matured lump-sum cohort restarts with $B_i = q \cdot \text{owed}_i$, $s_i = t$) (7)

Assumptions built into the ledger. New members are not owed anything in the month they join. Members never withdraw their principal from periodic schemes. Whatever is left in the pool at collapse goes

with the operators. With $g \le 1$, equation (2) keeps $E_t \le M$ and $D_t \ge 0$.

> **Accounting identity** (checked on every run): $E_C = \text{cash paid out} + k \cdot E_C + P'_C$ (8)

6. Why collapse is certain: the tipping point

Ignore the pool for a moment. For a monthly scheme, fresh cash covers the month's dues only while $(1 - k) \cdot D_t \ge r_m \cdot E$. Substituting (2):

> $(1 - k) \cdot g \cdot (1 - E/M) \ge r_m$
> $\Rightarrow \quad E^*/M = \mathbf{1 - r_m / ((1 - k) \cdot g)}$ (9)

Before E^*, the scheme takes in more than it pays and the pool grows. That growth is what makes it look healthy. After E^*, every month pays out more than comes in, and the scheme lives on its saved pool until the pool runs out. If $r_m \ge (1 - k)g$, then $E^* = 0$ and the scheme is underwater from the start. In plain terms: **once recruitment growth falls below the promised return, collapse is close.**

Because the model has no real business income, **any** positive return eventually collapses. That is why returns close to the bank rate get the SECP verdict (§10) instead of a date. At that level a real business could plausibly pay the return.

7. Where the family stands

> **Joining month:** $J = a$ if the scheme's age is known; otherwise $J = \min\{ t : E_t \ge E_C/2 \}$ (where the typical rupee enters) (10)
> **Months left:** $C - J \to$ Line 1 date = today + $(C - J)$ months (11)
> **Cash received:** $R = \Sigma\, A \cdot p$ over months $J+e, J+2e, \ldots$ strictly before C [lump sum: $R = A(1+r)$ if $J+e < C$, else 0] (12)
> **Loss:** $L = \max(0, A - R) \to$ Line 2 (13)
> **Bank profit:** $\Pi = A \cdot b \cdot (C - J)/12$ (simple profit, before tax) $\to$ Line 3 (14)

The family is assumed to take every payout as cash. If the known age $a \ge C$, the verdict is **overdue**.

8. Scheme-wide figures

> **Money lost:** $\Sigma_i \max(0, \text{put in}_i - \text{paid}_i) / \Sigma_i \text{put in}_i$ (15)
> **Money behind:** $\Sigma_i \text{put in}_i\, [\text{paid}_i < \text{put in}_i] / \Sigma_i \text{put in}_i$ (share of money whose owners end up with less than they put in) (16)

The calculator shows "money behind" as "X out of every 10 people". This treats each rupee as one person, which is exact only if all deposits are the same size.

8a. Scheme types

Recruitment differs by kind of scheme. An app spreads faster than a neighbourhood committee, and doubling schemes keep going because members re-invest. The user picks a type first; it fills a typical promise and replaces the defaults in §3.2 with that type's settings. Only "money doubling" is anchored to a real case (Double Shah: doubling in about 70 days, ran about two years). The others are placeholder judgements until calibrated.

Type	Typical promise (preset)	g	M	k	q	Preset lasts (months)
Something else (defaults)	—	0.3	1,000	0.1	0	—
Money doubling	+100% after 70 days	0.8	1,00,000	0.1	0.5	23
Committee doubling	+100% after 6 months	0.4	300	0.1	0.3	23
Crypto or app	2% daily	0.7	3,000	0.2	0.6	14
Forex trading	10% monthly	0.35	3,000	0.1	0.3	37
Modaraba or business	5% monthly	0.25	3,000	0.1	0.6	64

8b. Already invested

For a family already in a scheme, "now" is the scheme's age a if known, otherwise the typical joiner's month plus the months since they invested:

$N = a$ if known, else $J_{\text{typ}} + m_{\text{a}}$; joined at $J = \max(0,\ N - m_{\text{a}})$ (17)

Received so far: R_{u} if the family entered it, else $\Sigma\ A{\cdot}p$ over payout months in $(J,\ \bar{N}]$ before C (18)

At risk: $\max(0,\ A$ – received so far$)$; **months left:** $C - N$ (overdue if $N \geq C$) (19)

Bank, looking back: $\Pi = A \cdot b \cdot m_{\text{a}} / 12 \cdot (1 - \tau)$ (20)

8c. Promise in rupees, warning signs, read-aloud

Rupees per period: $r = P_{\text{Rs}} / A$ **Lump sum, total back:** $r = P_{\text{Rs}} / A - 1$ (must be > 0) (4a)

The warning-sign check asks six yes/no questions (guaranteed profit, bonus for recruiting, no SECP or State Bank licence shown, business not explained, pressure to decide fast, delays on withdrawal). 0 yes: no big signs, still check SECP. 1–2: warning. 3 or more: high risk. It does not change the model; it gives a second, number-free check, which matters most when the model gives the SECP verdict. A "Listen" button reads the result lines aloud with the phone's own text-to-speech voice (Urdu needs an Urdu voice installed). A chart shows the scheme's cash pool P_t from start to collapse, with "today" marked.

9. Verdict rules

Condition	Verdict	What is shown
$r_{\text{m}} \leq \sigma \cdot b_{\text{ref}}/12$	check-secp	No date. The user is told to check SECP permission; line 3 is still shown

No collapse within H months	check-secp	Same as above
Known age $a \geq C$	overdue	"By the numbers it should already have run out. Do not put in new money."
Otherwise	collapse	The three lines

10. Output lines (Urdu)

اس اسکیم کے پیسے زیادہ سے زیادہ {مہینہ سال} تک ختم ہو جائیں گے۔

Line 1: "This scheme's money will run out by {month year} at the latest." Followed by: "Most such schemes shut down before that."

اگر آپ {رقم} لگائیں تو تقریباً {نقصان} ڈوب جائیں گے۔

Line 2: "If you put in {amount}, about {loss} will be lost." If $L = 0$: "You may get your money back, but that profit will come from other people's lost money." Then: "X out of every 10 people in this scheme will lose. Late joiners lose almost everything."

یہی {رقم} {بینک} کے {اکاؤنٹ} میں {مہینے} مہینے رکھیں تو تقریباً {منافع} منافع ملے گا، اور اصل رقم بھی محفوظ رہے گی۔

Line 3: "Keep the same {amount} in {bank}'s {account} for {months} months and you'd earn about {profit}, and your original money stays safe."

یہ اسکیم پرانے ممبروں کو نئے ممبروں کے پیسوں سے منافع دیتی ہے۔ جب نئے لوگ آنا کم ہو جائیں گے تو پیسے ختم ہو جائیں گے۔

"Why?" tap: "This scheme pays old members with new members' money. When fewer new people join, the money will run out."

The calculator has an اردو / English switch; the English lines carry the same numbers (amounts written as Rs 1,50,000). Amounts are rounded and written in lakh/hazaar form (e.g. 1 لاکھ 50 ہزار). Months are written in Urdu (جنوری … دسمبر) with Western digits for the year.

11. Worked example: 10% a month, Rs 1 lakh

Defaults $g = 0.3$, $M = 1{,}000$, $k = 0.1$, $q = 0$. Values are in seed units. The amber row is the first month new cash (after the operators' cut) no longer covers dues. The red row is the collapse.

Month t	**New money** D_t	**Raised** E_t	**Into pool** $(1-k)D_t$	**Due** O_t	**Pool before paying** P'_t	**Pool after** P_t
0	1.000	1.000	0.900	0.000	0.900	0.900
1	0.300	1.300	0.270	0.100	1.170	1.070
2	0.389	1.689	0.350	0.130	1.420	1.290
3	0.506	2.195	0.455	0.169	1.746	1.577
4	0.657	2.852	0.591	0.219	2.168	1.948
5	0.853	3.705	0.768	0.285	2.716	2.431
⋮						
28	72.1	670	64.9	59.8	314	255

29	66.4	736	59.7	67.0	314	247
30	58.3	794	52.5	73.6	300	226
⋮						
32	39.6	883	35.7	84.3	226	142
33	31.0	914	27.9	88.3	170	81.8
34	23.6	938	21.2	91.4	103	11.6
35	**17.6**	**955**	**15.8**	**93.8**	**27.4**	**COLLAPSE: pool < due**

- Tipping point from (9): $E^*/M = 1 - 0.10 / (0.9 \times 0.3)$ = **63%** of the market. New cash first falls short of dues in month 29. The pool built up earlier carries the scheme until month **35**.
- Total raised by collapse: 955 seed units. Paid out in cash: 832. Kept by operators (cut + pool at collapse): 123.
- Typical joiner (J = month 26): 9 months left. Receives Rs 80,000 on Rs 1,00,000, so loses **Rs 20,000**. A bank at 10% would pay Rs 7,500 over the same months, and the principal stays safe.
- Scheme-wide: 29% of all money put in never comes back, and 60% of the money belongs to people who end up behind.

12. Results across scheme types

Defaults as above, Rs 1,00,000, bank 10% per year, typical joiner.

Scheme	Lasts (months)	Joins month	Months left	Gets back (Rs)	Loses (Rs)	Bank profit (Rs)	Money lost / behind
3% a month	56	26	30	87,000	13,000	25,000	15% / 79%
5% a month	44	26	18	85,000	15,000	15,000	20% / 69%
10% a month	35	26	9	80,000	20,000	7,500	29% / 60%
20% a month	26	23	3	40,000	60,000	2,500	49% / 68%
1% a day	9	7	2	30,417	69,583	1,667	55% / 73%
5% a week	25	22	3	43,452	56,548	2,500	51% / 71%
15% a quarter	45	26	19	90,000	10,000	15,833	21% / 69%
Double in 70 days	3	1	2	0	1,00,000	1,667	100% / 100%
+50% in 6 months	33	26	7	1,50,000	0	5,833	43% / 43%
2% a month	71	26	45	88,000	12,000	37,500	13% / 83%
1% a month	Verdict: check-secp (no date shown; at or below 1.5 × bank rate)						

Sensitivity: collapse month at 10% a month

g \ M	100	300	1,000	3,000	10,000

0.15	33	41	49	57	66
0.20	31	37	43	49	56
0.30	26	30	**35**	39	43
0.45	21	24	28	31	34
0.60	19	21	24	26	29

The date depends heavily on g and M. That is why calibration (§14) comes before any public release, and why line 1 says "at the latest" and adds that most schemes shut earlier.

Calibration probe: "double in 70 days" (Double Shah style)

g	q = 0		q = 0.5		q = 0.8	
	M=1,000	M=1,00,000	M=1,000	M=1,00,000	M=1,000	M=1,00,000
0.3	3	3	20	37	26	42
0.5	18	29	20	31	22	33
0.8	14	22	16	23	18	26

With the default g = 0.30 and no reinvestment, a doubling scheme collapses within 3 months. The real Double Shah scheme ran for roughly two years. The model reaches that only with faster recruitment, a larger market, or heavy reinvestment. So g, and probably q, should be set per type of scheme rather than globally.

13. Bank comparison data

Profit rates in % per year, before tax, for individuals with balances up to Rs 10 million. The data was checked on 2026-10-07. The SBP policy rate is 11.5%. Under SBP's 2026 minimum-profit rule for small savers, conventional banks show 10% on savings. Islamic rates are declared monthly pool returns and change every month.

Bank / institution	Savings (conventional)	Islamic savings	Fixed deposit / certificate, 1 year
National Savings (CDNS) قومی بچت (نیشنل سیونگز)	10.00% Savings Account, 2026-10	11.33% Sarwa Islamic Savings Account, 2026-10	11.31% Short Term Savings Certificate, 12 months, 2026-10
HBL ایچ بی ایل	10.00% PLS Savings, 2026-10	7.23% HBL Islamic savings (declared), 2026-06	10.60% HBL Advantage, 1 year, 2026-10
Meezan Bank میزان بینک	n/a	6.63% Savings (declared), 2026-09	9.56% Certificate of Islamic Investment, 1 year, 2026-09
UBL یو بی ایل	10.00% Savings Account, 2026-10	n/a	n/a
MCB ایم سی بی	10.00% Savings, 2026-08	n/a	7.12% Term Deposit, 1 year, 2026-08
Allied Bank الائیڈ بینک	10.00% PLS Savings, 2026-10	n/a	10.50% Behtar Munafa, 1 year, 2026-10

Bank Alfalah بینک الفلاح	10.00% PLS Savings, 2026-08	6.51% Alfalah Islamic savings, base (declared), 2026-06	10.20% Term Deposit below Rs 1m, 1 year, 2026-08
Bank AL Habib بینک الحبیب	10.00% Savings, 2026-10	8.00% Islamic savings (declared), 2026-06	10.00% Fixed Term, 1 year, 2026-08
JS Bank جے ایس بینک	10.00% Savings, 2026-10	n/a	10.90% Term Deposit, 1 year, 2026-10
BankIslami بینک اسلامی	n/a	6.78% Savings (declared), 2026-08	n/a
HabibMetro Sirat حبیب میٹرو (صراط)	n/a	8.48% Sirat savings (declared), 2026-06	n/a

- Line 3 applies the chosen annual rate as simple profit over the months left. A 1-year fixed deposit rate is pro-rated even when fewer than 12 months are left. This is an approximation, since early withdrawal may pay less.
- Withholding tax on profit (section 151) is applied as the user chooses: bank deposits 20% / 40%, National Savings 15% / 30% (filer / non-filer). Sources disagree (some give 15% / 30% for banks), so verify against the current Finance Act before release.
- National Savings Behbood certificates (12.72%) are available only to senior citizens, widows and persons with disabilities, so they are not used as a default.

Sources:

- Bank rate sheets as compiled by pakera.pk (savings and fixed deposit roundups, checked 6-7 Oct 2026)
- Islamic savings declarations as compiled by halalwallet.pk (June 2026)
- CDNS rate summary w.e.f. 1 Oct 2026, as compiled by pakera.pk

14. Limitations and calibration plan

1. **Not calibrated.** g, M, k, q are placeholders. The plan is to fit them to 3–5 collapsed schemes with a known promise and lifespan (Double Shah, the Modaraba scam, Pakistani crypto-app collapses) and to report the fit.
2. **"At the latest" is conditional.** Operators often run before the money runs out, which makes the real date earlier. Reinvestment ($q > 0$) makes the model date later. A date computed with $q = 0$ is therefore not a guaranteed upper bound.
3. **No withdrawals or bank runs.** Members pulling out principal, or a panic, would bring collapse earlier.
4. **No real income.** The model treats every payout as coming from new deposits. That is why returns near the bank rate get no date.
5. **Typical joiner.** Joining at the half-money point is a default for users who don't know the scheme's age. Late joiners lose much more than this figure.
6. **Monthly steps.** Daily and weekly payouts are summed by month. Lump-sum periods are rounded up to whole months.
7. **Bank rates go stale.** Rates are embedded in the offline file and must be refreshed and rebuilt periodically. The file shows the date the rates were checked.

15. References

- Artzrouni, M. (2009). The mathematics of Ponzi schemes. *Mathematical Social Sciences*, 58(2), 190–201.
- State Bank of Pakistan, monetary policy statements, 2026. Policy rate 11.5% (held 14 September 2026).

- Securities and Exchange Commission of Pakistan (SECP), secp.gov.pk: checking whether a company may take deposits from the public.

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